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Linear Systems of Simultaneous Equations

The previous chapter described a number of possible departures from the ideal regression model arising from errors that are non-Gaussian, heteroskedastic, or autocorrelated. We saw that while these factors can make a difference for the small-sample validity of t and F tests, under any of Assumptions 8.1 through 8.6, the OLS estimator $\mathbf{b}_T$ is either unbiased or consistent. This is because all these cases retained the crucial assumption that u_t , the error term for observation t , is uncorrelated with $\mathbf{x}_t$, the explanatory variables for that observation. Unfortunately, this critical assumption is unlikely to be satisfied in many important applications.

Section 9.1 discusses why this assumption often fails to hold, by examining a concrete example of *simultaneous equations bias*. Subsequent sections discuss a variety of techniques for dealing with this problem. These results will be used in the structural interpretation of vector autoregressions in Chapter 11 and for understanding generalized method of moments estimation in Chapter 14.

9.1. Simultaneous Equations Bias

To illustrate the difficulties with endogenous regressors, consider an investigation of the public's demand for oranges. Let p_t denote the log of the price of oranges in a particular year and q_t^d the log of the quantity the public is willing to buy. To keep the example very simple, suppose that price and quantity are covariance-stationary and that each is measured as deviations from its population mean. The demand curve is presumed to take the form

$$q_t^d = \beta p_t + \varepsilon_t^d, \quad [9.1.1]$$

with $\beta < 0$; a higher price reduces the quantity that the public is willing to buy. Here ε_t^d represents factors that influence demand other than price. These are assumed to be independent and identically distributed with mean zero and variance σ_d^2 .

The price also influences the supply of oranges brought to the market,

$$q_t^s = \gamma p_t + \varepsilon_t^s, \quad [9.1.2]$$

where $\gamma > 0$ and ε_t^s represents factors that influence supply other than price. These omitted factors are again assumed to be i.i.d. with mean zero and variance σ_s^2 , with the supply disturbance ε_t^s uncorrelated with the demand disturbance ε_t^d .

Equation [9.1.1] describes the behavior of buyers of oranges, and equation [9.1.2] describes the behavior of sellers. Market equilibrium requires $q_t^d = q_t^s$, or

$$\beta p_t + \varepsilon_t^d = \gamma p_t + \varepsilon_t^s.$$

Rearranging,

$$p_t = \frac{\varepsilon_t^d - \varepsilon_t^s}{\gamma - \beta}. \quad [9.1.3]$$

Substituting this back into [9.1.2],

$$q_t = \gamma \frac{\varepsilon_t^d - \varepsilon_t^s}{\gamma - \beta} + \varepsilon_t^s = \frac{\gamma}{\gamma - \beta} \varepsilon_t^d - \frac{\beta}{\gamma - \beta} \varepsilon_t^s. \quad [9.1.4]$$

Consider the consequences of trying to estimate [9.1.1] by *OLS*. A regression of quantity on price will produce the estimate

$$b_T = \frac{(1/T) \sum_{t=1}^T p_t q_t}{(1/T) \sum_{t=1}^T p_t^2}. \quad [9.1.5]$$

Substituting [9.1.3] and [9.1.4] into the numerator in [9.1.5] results in

$$\begin{aligned} \frac{1}{T} \sum_{t=1}^T p_t q_t &= \frac{1}{T} \sum_{t=1}^T \left[\frac{1}{\gamma - \beta} \varepsilon_t^d - \frac{1}{\gamma - \beta} \varepsilon_t^s \right] \left[\frac{\gamma}{\gamma - \beta} \varepsilon_t^d - \frac{\beta}{\gamma - \beta} \varepsilon_t^s \right] \\ &= \frac{1}{T} \sum_{t=1}^T \left[\frac{\gamma}{(\gamma - \beta)^2} (\varepsilon_t^d)^2 + \frac{\beta}{(\gamma - \beta)^2} (\varepsilon_t^s)^2 - \frac{\gamma + \beta}{(\gamma - \beta)^2} \varepsilon_t^d \varepsilon_t^s \right] \\ &\xrightarrow{p} \frac{\gamma \sigma_d^2 + \beta \sigma_s^2}{(\gamma - \beta)^2}. \end{aligned}$$

Similarly, for the denominator,

$$\frac{1}{T} \sum_{t=1}^T p_t^2 = \frac{1}{T} \sum_{t=1}^T \left[\frac{1}{\gamma - \beta} \varepsilon_t^d - \frac{1}{\gamma - \beta} \varepsilon_t^s \right]^2 \xrightarrow{p} \frac{\sigma_d^2 + \sigma_s^2}{(\gamma - \beta)^2}.$$

Hence,

$$b_T \xrightarrow{p} \left[\frac{\sigma_d^2 + \sigma_s^2}{(\gamma - \beta)^2} \right]^{-1} \left[\frac{\gamma \sigma_d^2 + \beta \sigma_s^2}{(\gamma - \beta)^2} \right] = \frac{\gamma \sigma_d^2 + \beta \sigma_s^2}{\sigma_d^2 + \sigma_s^2}. \quad [9.1.6]$$

OLS regression thus gives not the demand elasticity β but rather an average of β and the supply elasticity γ , with weights depending on the sizes of the variances σ_d^2 and σ_s^2 . If the error in the demand curve is negligible ($\sigma_d^2 \rightarrow 0$) or if the error term in the supply curve has a big enough variance ($\sigma_s^2 \rightarrow \infty$), then [9.1.6] indicates that *OLS* would give a consistent estimate of the demand elasticity β . On the other hand, if $\sigma_d^2 \rightarrow \infty$ or $\sigma_s^2 \rightarrow 0$, then *OLS* gives a consistent estimate of the supply elasticity γ . In the cases in between, one economist might believe the regression was estimating the demand curve [9.1.1] and a second economist might perform the same regression calling it the supply curve [9.1.2]. The actual *OLS* estimates would represent a mixture of both. This phenomenon is known as *simultaneous equations bias*.

Figure 9.1 depicts the problem graphically.¹ At any date in the sample, there is some demand curve (determined by the value of ε_t^d) and a supply curve (determined by ε_t^s), with the observation on (p_t, q_t) given by the intersection of these two curves. For example, date 1 may have been associated with a small negative shock to demand, producing the curve D_1 , and a large positive shock to supply, producing S_1 . The date 1 observation will then be (p_1, q_1) . Date 2 might have seen

¹Economists usually display these figures with the axes reversed from those displayed in Figure 9.1.

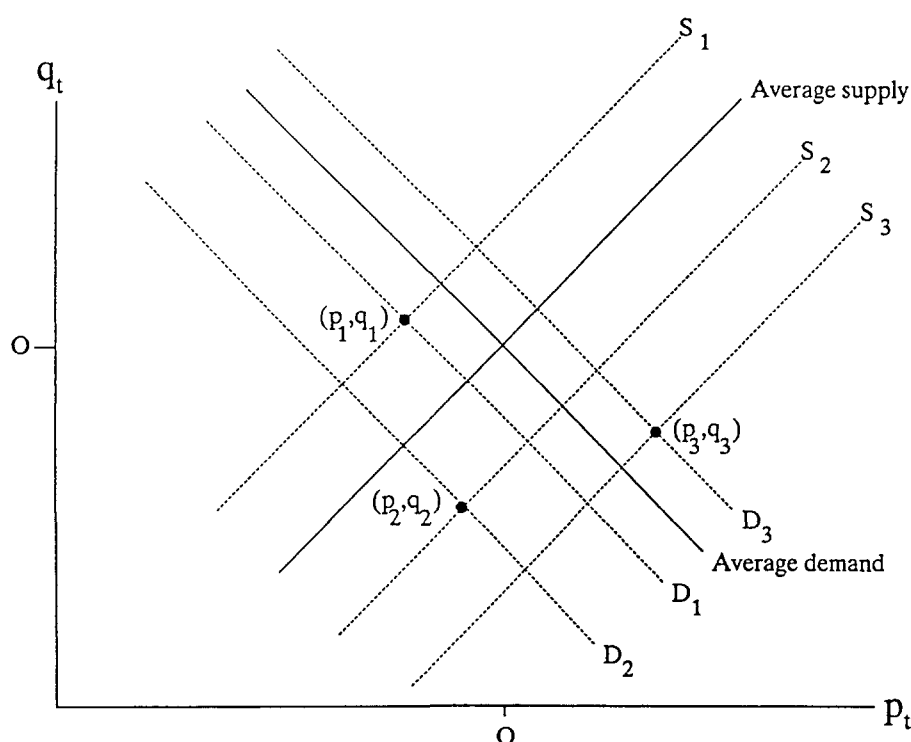


FIGURE 9.1 Observations on price and quantity implied by disturbances to both supply functions and demand functions.

a bigger negative shock to demand and a negative shock to supply, while date 3 as drawn reflects a modest positive shock to demand and a large negative shock to supply. *OLS* tries to fit a line through the scatter of points $\{p_t, q_t\}_{t=1}^T$.

If the shocks are known to be due to the supply curve and not the demand curve, then the scatter of points will trace out the demand curve, as in Figure 9.2. If the shocks are due to the demand curve rather than the supply curve, the scatter will trace out the supply curve, as in Figure 9.3.

The problem of simultaneous equations bias is extremely widespread in the social sciences. It is rare that the relation that we would like to estimate is the only possible reason why there might be a correlation among a group of variables.

Consistent Estimation of the Demand Elasticity

The above analysis suggests that consistent estimates of the demand elasticity might be obtained if we could find a variable that shifts the supply curve but not the demand curve. For example, let w_t represent the number of days of below-freezing temperatures in Florida during year t . Recalling that the supply disturbance ε_t^s was defined as factors influencing supply other than price, w_t seems likely to be an important component of ε_t^s . Define h to be the coefficient from a linear projection of ε_t^s on w_t , and write

$$\varepsilon_t^s = hw_t + u_t^s. \quad [9.1.7]$$

Thus, u_t^s is uncorrelated with w_t , by the definition of h . Although Florida weather is likely to influence the supply of oranges, it is natural to assume that weather

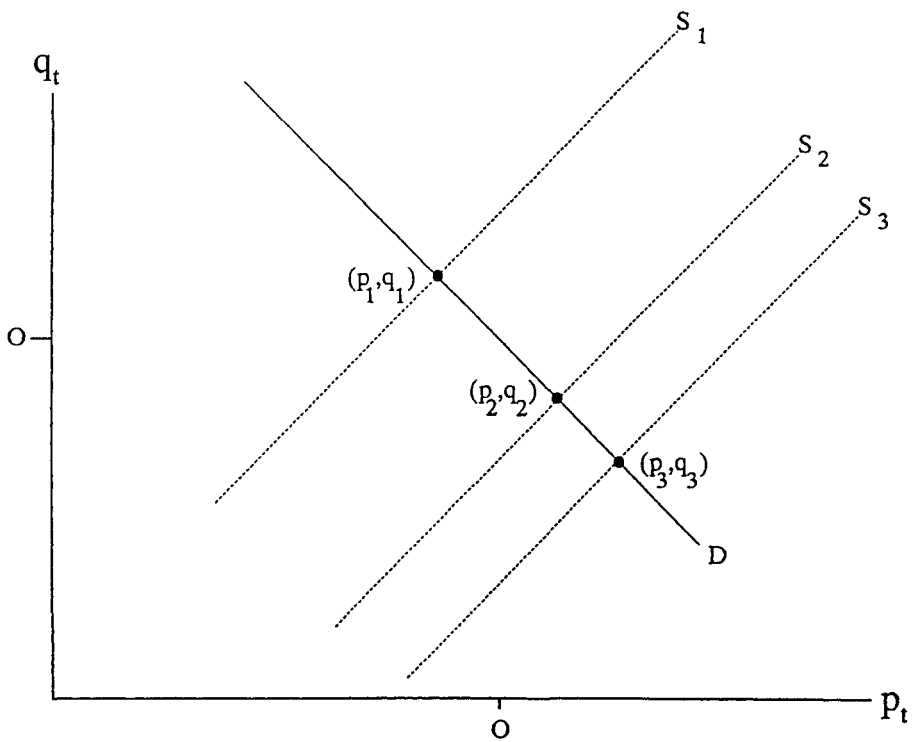


FIGURE 9.2 Observations on price and quantity implied by disturbances to supply function only.

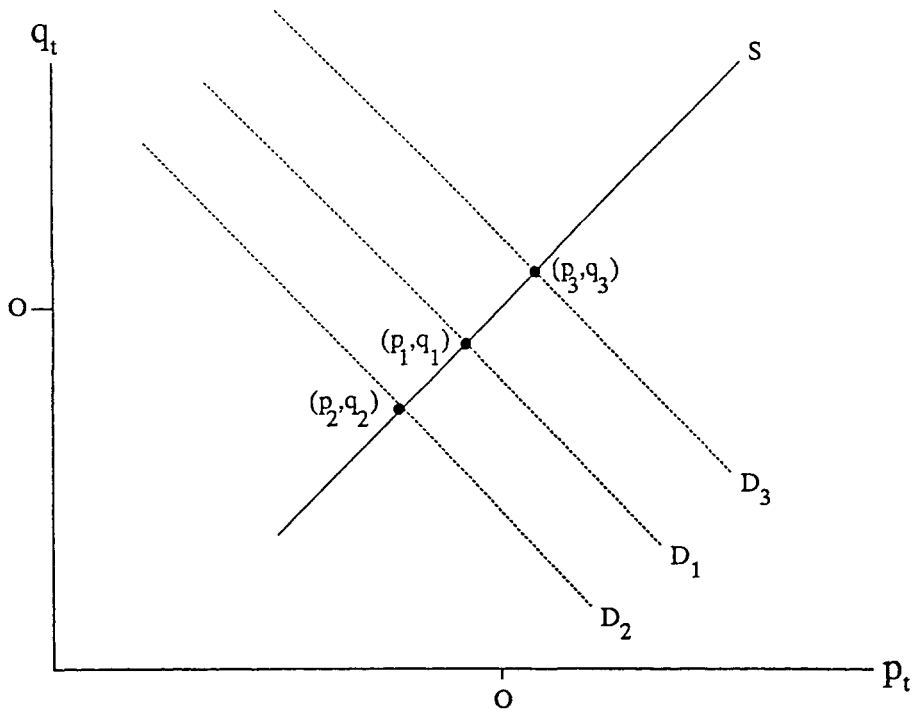


FIGURE 9.3 Observations on price and quantity implied by disturbances to demand function only.

matters for the public's demand for oranges only through its effect on the price. Under this assumption, both w_t and u_t^s are uncorrelated with ε_t^d . Changes in price that can be attributed to the weather represent supply shifts and not demand shifts.

Define p_t^* to be the linear projection of p_t on w_t . Substituting [9.1.7] into [9.1.3],

$$p_t = \frac{\varepsilon_t^d - h w_t - u_t^s}{\gamma - \beta}, \quad [9.1.8]$$

and thus

$$p_t^* = \frac{-h}{\gamma - \beta} w_t, \quad [9.1.9]$$

since ε_t^d and u_t^s are uncorrelated with w_t . Equation [9.1.8] can thus be written

$$p_t = p_t^* + \frac{\varepsilon_t^d - u_t^s}{\gamma - \beta},$$

and substituting this into [9.1.1],

$$q_t = \beta \left\{ p_t^* + \frac{\varepsilon_t^d - u_t^s}{\gamma - \beta} \right\} + \varepsilon_t^d = \beta p_t^* + v_t, \quad [9.1.10]$$

where

$$v_t \equiv \frac{-\beta u_t^s}{\gamma - \beta} + \frac{\gamma \varepsilon_t^d}{\gamma - \beta}.$$

Since u_t^s and ε_t^d are both uncorrelated with w_t , it follows that v_t is uncorrelated with p_t^* . Hence, if [9.1.10] were estimated by ordinary least squares, the result would be a consistent estimate of β :

$$\begin{aligned} \hat{\beta}_T^* &= \frac{(1/T) \sum_{t=1}^T p_t^* q_t}{(1/T) \sum_{t=1}^T [p_t^*]^2} \\ &= \frac{(1/T) \sum_{t=1}^T p_t^* (\beta p_t^* + v_t)}{(1/T) \sum_{t=1}^T [p_t^*]^2} \\ &= \beta + \frac{(1/T) \sum_{t=1}^T p_t^* v_t}{(1/T) \sum_{t=1}^T [p_t^*]^2} \\ &\xrightarrow{p} \beta. \end{aligned} \quad [9.1.11]$$

The suggestion is thus to regress quantity on that component of price that is induced by the weather, that is, regress quantity on the linear projection of price on the weather.

In practice, we will not know the values of the population parameters h , γ , and β necessary to construct p_t^* in [9.1.9]. However, the linear projection p_t^* can be consistently estimated by the fitted value for observation t from an *OLS* regression of p on w ,

$$\hat{p}_t = \hat{\delta}_T w_t, \quad [9.1.12]$$

where

$$\hat{\delta}_T = \frac{(1/T) \sum_{t=1}^T w_t p_t}{(1/T) \sum_{t=1}^T w_t^2}.$$

The estimator [9.1.11] with p_t^* replaced by $\hat{p}_t$ is known as the *two-stage least squares* (2SLS) coefficient estimator:

$$\hat{\beta}_{2SLS} = \frac{(1/T) \sum_{t=1}^T \hat{p}_t q_t}{(1/T) \sum_{t=1}^T [\hat{p}_t]^2}. \quad [9.1.13]$$

Like $\hat{\beta}_T^*$, the 2SLS estimator is consistent, as will be shown in the following section.

9.2. Instrumental Variables and Two-Stage Least Squares

General Description of Two-Stage Least Squares

A generalization of the previous example is as follows. Suppose the objective is to estimate the vector β in the regression model

$$y_t = \beta' z_t + u_t, \quad [9.2.1]$$

where z_t is a $(k \times 1)$ vector of explanatory variables. Some subset $n \leq k$ of the variables in z_t are thought to be endogenous, that is, correlated with u_t . The remaining $k - n$ variables in z_t are said to be *predetermined*, meaning that they are uncorrelated with u_t . Estimation of β requires variables known as *instruments*. To be a valid instrument, a variable must be correlated with an endogenous explanatory variable in z_t but uncorrelated with the regression disturbance u_t . In the supply-and-demand example, the weather variable w_t served as an instrument for price. At least one valid instrument must be found for each endogenous explanatory variable.

Collect the predetermined explanatory variables together with the instruments in an $(r \times 1)$ vector x_t . For example, to estimate the demand curve, there were no predetermined explanatory variables in equation [9.1.1] and only a single instrument; hence, $r = 1$, and x_t would be the scalar w_t . As a second example, suppose that the equation to be estimated is

$$y_t = \beta_1 + \beta_2 z_{2t} + \beta_3 z_{3t} + \beta_4 z_{4t} + \beta_5 z_{5t} + u_t.$$

In this example, z_{4t} and z_{5t} are endogenous (meaning that they are correlated with u_t), z_{2t} and z_{3t} are predetermined (uncorrelated with u_t), and ξ_{1t} , ξ_{2t} , and ξ_{3t} are valid instruments (correlated with z_{4t} and z_{5t} but uncorrelated with u_t). Then $r = 6$ and $x_t' = (1, z_{2t}, z_{3t}, \xi_{1t}, \xi_{2t}, \xi_{3t})$. The requirement that there be at least as many instruments as endogenous explanatory variables implies that $r \geq k$.

Consider an *OLS* regression of z_{it} (the i th explanatory variable in [9.2.1]) on x_t :

$$z_{it} = \delta_i' x_t + e_{it}. \quad [9.2.2]$$

The fitted values for the regression are given by

$$\hat{z}_{it} = \hat{\delta}_i' x_t, \quad [9.2.3]$$

where

$$\hat{\boldsymbol{\delta}}_i = \left[\sum_{t=1}^T \mathbf{x}_t \mathbf{x}_t' \right]^{-1} \left[\sum_{t=1}^T \mathbf{x}_t z_{it} \right].$$

If z_{it} is one of the predetermined variables, then z_{it} is one of the elements of $\mathbf{x}_t$, and equation [9.2.3] simplifies to

$$\hat{z}_{it} = z_{it}.$$

This is because when the dependent variable (z_{it}) is included in the regressors ($\mathbf{x}_t$), a unit coefficient on z_{it} and zero coefficients on the other variables produce a perfect fit and thus minimize the residual sum of squares.

Collect the equations in [9.2.3] for $i = 1, 2, \dots, k$ in a $(k \times 1)$ vector equation

$$\hat{\mathbf{z}}_t = \hat{\boldsymbol{\delta}}' \mathbf{x}_t, \quad [9.2.4]$$

where the $(k \times r)$ matrix $\hat{\boldsymbol{\delta}}'$ is given by

$$\hat{\boldsymbol{\delta}}' = \begin{bmatrix} \hat{\boldsymbol{\delta}}_1' \\ \hat{\boldsymbol{\delta}}_2' \\ \vdots \\ \hat{\boldsymbol{\delta}}_k' \end{bmatrix} = \left[\sum_{t=1}^T \mathbf{z}_t \mathbf{x}_t' \right] \left[\sum_{t=1}^T \mathbf{x}_t \mathbf{x}_t' \right]^{-1}. \quad [9.2.5]$$

The *two-stage least squares* (2SLS) estimate of $\boldsymbol{\beta}$ is found from an *OLS* regression of y_t on $\hat{\mathbf{z}}_t$:

$$\hat{\boldsymbol{\beta}}_{2SLS} = \left[\sum_{t=1}^T \hat{\mathbf{z}}_t \hat{\mathbf{z}}_t' \right]^{-1} \left[\sum_{t=1}^T \hat{\mathbf{z}}_t y_t \right]. \quad [9.2.6]$$

An alternative way of writing [9.2.6] is sometimes useful. Let $\hat{e}_{it}$ denote the sample residual from *OLS* estimation of [9.2.2]; that is, let

$$z_{it} = \hat{\boldsymbol{\delta}}_i' \mathbf{x}_t + \hat{e}_{it} = \hat{z}_{it} + \hat{e}_{it}. \quad [9.2.7]$$

OLS causes this residual to be orthogonal to $\mathbf{x}_t$:

$$\sum_{t=1}^T \mathbf{x}_t \hat{e}_{it} = \mathbf{0},$$

meaning that the residual is orthogonal to $\hat{z}_{jt}$:

$$\sum_{t=1}^T \hat{z}_{jt} \hat{e}_{it} = \hat{\boldsymbol{\delta}}_j' \sum_{t=1}^T \mathbf{x}_t \hat{e}_{it} = 0.$$

Hence, if [9.2.7] is multiplied by $\hat{z}_{jt}$ and summed over t , the result is

$$\sum_{t=1}^T \hat{z}_{jt} z_{it} = \sum_{t=1}^T \hat{z}_{jt} (\hat{z}_{it} + \hat{e}_{it}) = \sum_{t=1}^T \hat{z}_{jt} \hat{z}_{it}$$

for all i and j . This means that

$$\sum_{t=1}^T \hat{\mathbf{z}}_t \mathbf{z}_t' = \sum_{t=1}^T \hat{\mathbf{z}}_t \hat{\mathbf{z}}_t',$$

so that the 2SLS estimator [9.2.6] can equivalently be written as

$$\hat{\boldsymbol{\beta}}_{2SLS} = \left[\sum_{t=1}^T \hat{\mathbf{z}}_t \mathbf{z}_t' \right]^{-1} \left[\sum_{t=1}^T \hat{\mathbf{z}}_t y_t \right]. \quad [9.2.8]$$

Consistency of 2SLS Estimator

Substituting [9.2.1] into [9.2.8],

$$\begin{aligned}\hat{\boldsymbol{\beta}}_{2SLS,T} &= \left[\sum_{t=1}^T \hat{\mathbf{z}}_t \mathbf{z}'_t \right]^{-1} \left[\sum_{t=1}^T \hat{\mathbf{z}}_t (\mathbf{z}'_t \boldsymbol{\beta} + u_t) \right] \\ &= \boldsymbol{\beta} + \left[\sum_{t=1}^T \hat{\mathbf{z}}_t \mathbf{z}'_t \right]^{-1} \left[\sum_{t=1}^T \hat{\mathbf{z}}_t u_t \right],\end{aligned}\quad [9.2.9]$$

where the subscript T has been added to keep explicit track of the sample size T on which estimation is based. It follows from [9.2.9] that

$$\hat{\boldsymbol{\beta}}_{2SLS,T} - \boldsymbol{\beta} = \left[(1/T) \sum_{t=1}^T \hat{\mathbf{z}}_t \mathbf{z}'_t \right]^{-1} \left[(1/T) \sum_{t=1}^T \hat{\mathbf{z}}_t u_t \right]. \quad [9.2.10]$$

Consistency of the 2SLS estimator can then be shown as follows. First note from [9.2.4] and [9.2.5] that

$$\begin{aligned}(1/T) \sum_{t=1}^T \hat{\mathbf{z}}_t \mathbf{z}'_t &= \hat{\boldsymbol{\delta}}'_T (1/T) \sum_{t=1}^T \mathbf{x}_t \mathbf{z}'_t \\ &= \left[(1/T) \sum_{t=1}^T \mathbf{z}_t \mathbf{x}'_t \right] \left[(1/T) \sum_{t=1}^T \mathbf{x}_t \mathbf{x}'_t \right]^{-1} \left[(1/T) \sum_{t=1}^T \mathbf{x}_t \mathbf{z}'_t \right].\end{aligned}\quad [9.2.11]$$

Assuming that the process $(\mathbf{z}_t, \mathbf{x}_t)$ is covariance-stationary and ergodic for second moments,

$$(1/T) \sum_{t=1}^T \hat{\mathbf{z}}_t \mathbf{z}'_t \xrightarrow{p} \mathbf{Q}, \quad [9.2.12]$$

where

$$\mathbf{Q} = [E(\mathbf{z}_t \mathbf{x}'_t)][E(\mathbf{x}_t \mathbf{x}'_t)]^{-1}[E(\mathbf{x}_t \mathbf{z}'_t)]. \quad [9.2.13]$$

Turning next to the second term in [9.2.10],

$$\left[(1/T) \sum_{t=1}^T \hat{\mathbf{z}}_t u_t \right] = \hat{\boldsymbol{\delta}}'_T (1/T) \sum_{t=1}^T \mathbf{x}_t u_t.$$

Again, ergodicity for second moments implies from [9.2.5] that

$$\hat{\boldsymbol{\delta}}'_T \xrightarrow{p} [E(\mathbf{z}_t \mathbf{x}'_t)][E(\mathbf{x}_t \mathbf{x}'_t)]^{-1}, \quad [9.2.14]$$

while the law of large numbers will typically ensure that

$$(1/T) \sum_{t=1}^T \mathbf{x}_t u_t \xrightarrow{p} E(\mathbf{x}_t u_t) = \mathbf{0},$$

under the assumed absence of correlation between $\mathbf{x}_t$ and u_t . Hence,

$$\left[(1/T) \sum_{t=1}^T \hat{\mathbf{z}}_t u_t \right] \xrightarrow{p} \mathbf{0}. \quad [9.2.15]$$

Substituting [9.2.12] and [9.2.15] into [9.2.10], it follows that

$$\hat{\boldsymbol{\beta}}_{2SLS,T} - \boldsymbol{\beta} \xrightarrow{p} \mathbf{Q}^{-1} \cdot \mathbf{0} = \mathbf{0}.$$

Hence, the 2SLS estimator is consistent as long as the matrix $\mathbf{Q}$ in [9.2.13] is nonsingular.

Notice that if none of the predetermined variables is correlated with z_{it} , then the i th row of $E(\mathbf{z}_i \mathbf{x}_i')$ contains all zeros and the corresponding row of $\mathbf{Q}$ in [9.2.13] contains all zeros, in which case 2SLS is not consistent. Alternatively, if z_{it} is correlated with $\mathbf{x}_i$ only through, say, the first element x_{1i} and z_{jt} is also correlated with $\mathbf{x}_i$ only through x_{1i} , then subtracting some multiple of the i th row of $\mathbf{Q}$ from the j th row produces a row of zeros, and $\mathbf{Q}$ again is not invertible. In general, consistency of the 2SLS estimator requires the rows of $E(\mathbf{z}_i \mathbf{x}_i')$ to be linearly independent. This essentially amounts to the requirement that there be a way of assigning instruments to endogenous variables such that each endogenous variable has an instrument associated with it, with no instrument counted twice for this purpose.

Asymptotic Distribution of 2SLS Estimator

Equation [9.2.10] implies that

$$\sqrt{T}(\hat{\boldsymbol{\beta}}_{2SLS,T} - \boldsymbol{\beta}) = \left[(1/T) \sum_{t=1}^T \hat{\mathbf{z}}_t \mathbf{z}_t' \right]^{-1} \left[(1/\sqrt{T}) \sum_{t=1}^T \hat{\mathbf{z}}_t u_t \right], \quad [9.2.16]$$

where

$$\left[(1/\sqrt{T}) \sum_{t=1}^T \hat{\mathbf{z}}_t u_t \right] = \hat{\boldsymbol{\delta}}_T' (1/\sqrt{T}) \sum_{t=1}^T \mathbf{x}_t u_t.$$

Hence, from [9.2.12] and [9.2.14],

$$\sqrt{T}(\hat{\boldsymbol{\beta}}_{2SLS,T} - \boldsymbol{\beta}) \xrightarrow{P} \mathbf{Q}^{-1} [E(\mathbf{z}_i \mathbf{x}_i')] [E(\mathbf{x}_i \mathbf{x}_i')]^{-1} \left((1/\sqrt{T}) \sum_{t=1}^T \mathbf{x}_t u_t \right). \quad [9.2.17]$$

Suppose that $\mathbf{x}_i$ is covariance-stationary and that $\{u_i\}$ is an i.i.d. sequence with mean zero and variance σ^2 with u_i independent of $\mathbf{x}_s$ for all $s \leq t$. Then $\{\mathbf{x}_i u_i\}$ is a martingale difference sequence with variance-covariance matrix given by $\sigma^2 \cdot E(\mathbf{x}_i \mathbf{x}_i')$. If u_i and $\mathbf{x}_i$ have finite fourth moments, then we can expect from Proposition 7.9 that

$$\left((1/\sqrt{T}) \sum_{t=1}^T \mathbf{x}_t u_t \right) \xrightarrow{L} N(\mathbf{0}, \sigma^2 \cdot E(\mathbf{x}_i \mathbf{x}_i')). \quad [9.2.18]$$

Thus, [9.2.17] implies that

$$\sqrt{T}(\hat{\boldsymbol{\beta}}_{2SLS,T} - \boldsymbol{\beta}) \xrightarrow{L} N(\mathbf{0}, \mathbf{V}), \quad [9.2.19]$$

where

$$\begin{aligned} \mathbf{V} &= \mathbf{Q}^{-1} [E(\mathbf{z}_i \mathbf{x}_i')] [E(\mathbf{x}_i \mathbf{x}_i')]^{-1} [\sigma^2 \cdot E(\mathbf{x}_i \mathbf{x}_i')] [E(\mathbf{x}_i \mathbf{x}_i')]^{-1} [E(\mathbf{x}_i \mathbf{z}_i')] \mathbf{Q}^{-1} \\ &= \sigma^2 \mathbf{Q}^{-1} \cdot \mathbf{Q} \cdot \mathbf{Q}^{-1} \\ &= \sigma^2 \mathbf{Q}^{-1} \end{aligned} \quad [9.2.20]$$

for $\mathbf{Q}$ given in [9.2.13]. Hence,

$$\hat{\boldsymbol{\beta}}_{2SLS,T} \approx N(\boldsymbol{\beta}, (1/T) \sigma^2 \mathbf{Q}^{-1}). \quad [9.2.21]$$

9.2. Instrumental Variables and Two-Stage Least Squares **241**

Since $\hat{\beta}_{2SLS}$ is a consistent estimate of β , clearly a consistent estimate of the population residual for observation t is afforded by

$$\hat{u}_{t,T} \equiv y_t - \mathbf{z}_t' \hat{\beta}_{2SLS,T} \xrightarrow{p} u_t. \quad [9.2.22]$$

Similarly, it is straightforward to show that σ^2 can be consistently estimated by

$$\hat{\sigma}_T^2 = (1/T) \sum_{t=1}^T (y_t - \mathbf{z}_t' \hat{\beta}_{2SLS,T})^2 \quad [9.2.23]$$

(see Exercise 9.1). Note well that although $\hat{\beta}_{2SLS}$ can be calculated from an *OLS* regression of y_t on $\hat{\mathbf{z}}_t$, the estimates $\hat{u}_t$ and $\hat{\sigma}^2$ in [9.2.22] and [9.2.23] are not based on the residuals from this regression:

$$\begin{aligned} \hat{u}_t &\neq y_t - \hat{\mathbf{z}}_t' \hat{\beta}_{2SLS} \\ \hat{\sigma}^2 &\neq (1/T) \sum_{t=1}^T (y_t - \hat{\mathbf{z}}_t' \hat{\beta}_{2SLS})^2. \end{aligned}$$

The correct estimates [9.2.22] and [9.2.23] use the actual explanatory variables $\mathbf{z}_t$ rather than the fitted values $\hat{\mathbf{z}}_t$.

A consistent estimate of $\mathbf{Q}$ is provided by [9.2.11]:

$$\begin{aligned} \hat{\mathbf{Q}}_T &= (1/T) \sum_{t=1}^T \hat{\mathbf{z}}_t \hat{\mathbf{z}}_t' \\ &= \left[(1/T) \sum_{t=1}^T \mathbf{z}_t \mathbf{x}_t' \right] \left[(1/T) \sum_{t=1}^T \mathbf{x}_t \mathbf{x}_t' \right]^{-1} \left[(1/T) \sum_{t=1}^T \mathbf{x}_t \mathbf{z}_t' \right]. \end{aligned} \quad [9.2.24]$$

Substituting [9.2.23] and [9.2.24] into [9.2.21], the estimated variance-covariance matrix of the 2SLS estimator is

$$\begin{aligned} \hat{\mathbf{V}}_T/T &= \hat{\sigma}_T^2 \cdot (1/T) \cdot \left[(1/T) \sum_{t=1}^T \hat{\mathbf{z}}_t \hat{\mathbf{z}}_t' \right]^{-1} \\ &= \hat{\sigma}_T^2 \cdot \left\{ \left[\sum_{t=1}^T \mathbf{z}_t \mathbf{x}_t' \right] \left[\sum_{t=1}^T \mathbf{x}_t \mathbf{x}_t' \right]^{-1} \left[\sum_{t=1}^T \mathbf{x}_t \mathbf{z}_t' \right] \right\}^{-1}. \end{aligned} \quad [9.2.25]$$

A test of the null hypothesis $\mathbf{R}\beta = \mathbf{r}$ can thus be based on

$$(\mathbf{R}\hat{\beta}_{2SLS,T} - \mathbf{r})' [\mathbf{R}(\hat{\mathbf{V}}_T/T)\mathbf{R}']^{-1} (\mathbf{R}\hat{\beta}_{2SLS,T} - \mathbf{r}), \quad [9.2.26]$$

which, under the null hypothesis, has an asymptotic distribution that is χ^2 with degrees of freedom given by m , where m represents the number of restrictions or the number of rows of $\mathbf{R}$.

Heteroskedasticity- and autocorrelation-consistent standard errors for 2SLS estimation will be discussed in Chapter 14.

Instrumental Variable Estimation

Substituting [9.2.4] and [9.2.5] into [9.2.8], the 2SLS estimator can be written as

$$\begin{aligned} \hat{\beta}_{2SLS} &= \left[\sum_{t=1}^T \hat{\delta}'_t \mathbf{x}_t \mathbf{z}_t' \right]^{-1} \left[\sum_{t=1}^T \hat{\delta}'_t \mathbf{x}_t y_t \right] \\ &= \left\{ \left[\sum_{t=1}^T \mathbf{z}_t \mathbf{x}_t' \right] \left[\sum_{t=1}^T \mathbf{x}_t \mathbf{x}_t' \right]^{-1} \left[\sum_{t=1}^T \mathbf{x}_t \mathbf{z}_t' \right] \right\}^{-1} \left\{ \left[\sum_{t=1}^T \mathbf{z}_t \mathbf{x}_t' \right] \left[\sum_{t=1}^T \mathbf{x}_t \mathbf{x}_t' \right]^{-1} \left[\sum_{t=1}^T \mathbf{x}_t y_t \right] \right\}. \end{aligned} \quad [9.2.27]$$

Consider the special case in which the number of instruments is exactly equal to the number of endogenous explanatory variables, so that $r = k$, as was the case for estimation of the demand curve in Section 9.1. Then $\sum_{t=1}^T \mathbf{z}_t \mathbf{x}_t'$ is a $(k \times k)$ matrix and [9.2.27] becomes

$$\begin{aligned} \hat{\boldsymbol{\beta}}_{IV} &= \left\{ \left[\sum_{t=1}^T \mathbf{x}_t \mathbf{z}_t' \right]^{-1} \left[\sum_{t=1}^T \mathbf{x}_t \mathbf{x}_t' \right] \left[\sum_{t=1}^T \mathbf{z}_t \mathbf{x}_t' \right]^{-1} \right\} \\ &\quad \times \left\{ \left[\sum_{t=1}^T \mathbf{z}_t \mathbf{x}_t' \right] \left[\sum_{t=1}^T \mathbf{x}_t \mathbf{x}_t' \right]^{-1} \left[\sum_{t=1}^T \mathbf{x}_t y_t \right] \right\} \\ &= \left[\sum_{t=1}^T \mathbf{x}_t \mathbf{z}_t' \right]^{-1} \left[\sum_{t=1}^T \mathbf{x}_t y_t \right]. \end{aligned} \quad [9.2.28]$$

Expression [9.2.28] is known as the *instrumental variable (IV)* estimator.

A key property of the *IV* estimator can be seen by premultiplying both sides of [9.2.28] by $\sum_{t=1}^T \mathbf{x}_t \mathbf{z}_t'$:

$$\sum_{t=1}^T \mathbf{x}_t \mathbf{z}_t' \hat{\boldsymbol{\beta}}_{IV} = \sum_{t=1}^T \mathbf{x}_t y_t,$$

implying that

$$\sum_{t=1}^T \mathbf{x}_t (y_t - \mathbf{z}_t' \hat{\boldsymbol{\beta}}_{IV}) = \mathbf{0}. \quad [9.2.29]$$

Thus, the *IV* sample residual $(y_t - \mathbf{z}_t' \hat{\boldsymbol{\beta}}_{IV})$ has the property that it is orthogonal to the instruments $\mathbf{x}_t$, in contrast to the *OLS* sample residual $(y_t - \mathbf{z}_t' \mathbf{b})$, which is orthogonal to the explanatory variables $\mathbf{z}_t$. The *IV* estimator is preferred to *OLS* because the population residual of the equation we are trying to estimate (u_t) is correlated with $\mathbf{z}_t$ but uncorrelated with $\mathbf{x}_t$.

Since the *IV* estimator is a special case of *2SLS*, it shares the consistency property of the *2SLS* estimator. Its estimated variance with i.i.d. residuals can be calculated from [9.2.25]:

$$\hat{\sigma}_T^2 \left[\sum_{t=1}^T \mathbf{x}_t \mathbf{z}_t' \right]^{-1} \left[\sum_{t=1}^T \mathbf{x}_t \mathbf{x}_t' \right] \left[\sum_{t=1}^T \mathbf{z}_t \mathbf{x}_t' \right]^{-1}. \quad [9.2.30]$$

9.3. Identification

We noted in the supply-and-demand example in Section 9.1 that the demand elasticity β could not be estimated consistently by an *OLS* regression of quantity on price. Indeed, in the absence of a valid instrument such as w_t , the demand elasticity cannot be estimated by any method! To see this, recall that the system as written in [9.1.1] and [9.1.2] implied the expressions [9.1.4] and [9.1.3]:

$$\begin{aligned} q_t &= \frac{\gamma}{\gamma - \beta} \varepsilon_t^d - \frac{\beta}{\gamma - \beta} \varepsilon_t^s \\ p_t &= \frac{\varepsilon_t^d - \varepsilon_t^s}{\gamma - \beta}. \end{aligned}$$

If ε_t^d and ε_t^s are i.i.d. Gaussian, then these equations imply that the vector $(q_t, p_t)'$ is Gaussian with mean zero and variance-covariance matrix

$$\Omega \equiv [1/(\gamma - \beta)^2] \cdot \begin{bmatrix} \gamma^2 \sigma_d^2 + \beta^2 \sigma_s^2 & \gamma \sigma_d^2 + \beta \sigma_s^2 \\ \gamma \sigma_d^2 + \beta \sigma_s^2 & \sigma_d^2 + \sigma_s^2 \end{bmatrix}.$$

This matrix is completely described by three magnitudes, these being the variances of q and p along with their covariance. Given a large enough sample, the values of these three magnitudes can be inferred with considerable confidence, but that is all that can be inferred, because these magnitudes can completely specify the process that generated the data under the maintained assumption of zero-mean i.i.d. Gaussian observations. There is no way to uncover the four parameters of the structural model $(\beta, \gamma, \sigma_d^2, \sigma_s^2)$ from these three magnitudes. For example, the values $(\beta, \gamma, \sigma_d^2, \sigma_s^2) = (1, 2, 3, 4)$ imply exactly the same observable properties for the data as would $(\beta, \gamma, \sigma_d^2, \sigma_s^2) = (2, 1, 4, 3)$.

If two different values for a parameter vector θ imply the same probability distribution for the observed data, then the vector θ is said to be *unidentified*.

When a third Gaussian white noise variable w_t is added to the set of observations, three additional magnitudes are available to characterize the process for observables, these being the variance of w , the covariance between w and p , and the covariance between w and q . If the new variable w enters both the demand and the supply equation, then three new parameters would be required to estimate the structural model—the parameter that summarizes the effect of w on demand, the parameter that summarizes its effect on supply, and the variance of w . With three more estimable magnitudes but three more parameters to estimate, we would be stuck with the same problem, having no basis for estimation of β .

Consistent estimation of the demand elasticity was achieved by using two-stage least squares because it was assumed that w appeared in the supply equation but was excluded from the demand equation. This is known as achieving identification through *exclusion restrictions*.

We showed in Section 9.2 that the parameters of an equation could be estimated (and thus must be identified) if (1) the number of instruments for that equation is at least as great as the number of endogenous explanatory variables for that equation and (2) the rows of $E(\mathbf{z}_t \mathbf{x}_t')$ are linearly independent. The first condition is known as the *order* condition for identification, and the second is known as the *rank* condition.

The rank condition for identification can be summarized more explicitly by specifying a complete system of equations for all of the endogenous variables. Let $\mathbf{y}_t$ denote an $(n \times 1)$ vector containing all of the endogenous variables in the system, and let $\mathbf{x}_t$ denote an $(m \times 1)$ vector containing all of the predetermined variables. Suppose that the system consists of n equations written as

$$\mathbf{B} \mathbf{y}_t + \Gamma \mathbf{x}_t = \mathbf{u}_t, \quad [9.3.1]$$

where $\mathbf{B}$ and Γ are $(n \times n)$ and $(n \times m)$ matrices of coefficients, respectively, and $\mathbf{u}_t$ is an $(n \times 1)$ vector of disturbances. The statement that $\mathbf{x}_t$ is predetermined is taken to mean that $E(\mathbf{x}_t \mathbf{u}_t') = \mathbf{0}$. For example, the demand and supply equations considered in Section 9.1 were

$$q_t = \beta p_t + u_t^d \quad (\text{demand}) \quad [9.3.2]$$

$$q_t = \gamma p_t + h w_t + u_t^s \quad (\text{supply}). \quad [9.3.3]$$

For this system, there are $n = 2$ endogenous variables, with $\mathbf{y}_t = (q_t, p_t)'$; and $m = 1$ predetermined variable, so that $\mathbf{x}_t = w_t$. This system can be written in the form of [9.3.1] as

$$\begin{bmatrix} 1 & -\beta \\ 1 & -\gamma \end{bmatrix} \begin{bmatrix} q_t \\ p_t \end{bmatrix} + \begin{bmatrix} 0 \\ -h \end{bmatrix} w_t = \begin{bmatrix} u_t^d \\ u_t^s \end{bmatrix}. \quad [9.3.4]$$

Suppose we are interested in the equation represented by the first row of the vector system of equations in [9.3.1]. Let y_{0t} be the dependent variable in the first equation, and let $\mathbf{y}_{1t}$ denote an $(n_1 \times 1)$ vector consisting of those endogenous variables that appear in the first equation as explanatory variables. Similarly, let $\mathbf{x}_{1t}$ denote an $(m_1 \times 1)$ vector consisting of those predetermined variables that appear in the first equation as explanatory variables. Then the first equation in the system is

$$y_{0t} + \mathbf{B}_{01}\mathbf{y}_{1t} + \mathbf{\Gamma}_{01}\mathbf{x}_{1t} = u_{0t},$$

where $\mathbf{B}_{01}$ is a $(1 \times n_1)$ vector and $\mathbf{\Gamma}_{01}$ is a $(1 \times m_1)$ vector. Let $\mathbf{y}_{2t}$ denote an $(n_2 \times 1)$ vector consisting of those endogenous variables that do not appear in the first equation; thus, $\mathbf{y}'_t = (y_{0t}, \mathbf{y}'_{1t}, \mathbf{y}'_{2t})$ and $1 + n_1 + n_2 = n$. Similarly, let $\mathbf{x}_{2t}$ denote an $(m_2 \times 1)$ vector consisting of those predetermined variables that do not appear in the first equation, so that $\mathbf{x}'_t = (\mathbf{x}'_{1t}, \mathbf{x}'_{2t})$ and $m_1 + m_2 = m$. Then the system in [9.3.1] can be written in partitioned form as

$$\begin{bmatrix} 1 & \mathbf{B}_{01} & \mathbf{0}' \\ \mathbf{B}_{10} & \mathbf{B}_{11} & \mathbf{B}_{12} \\ \mathbf{B}_{20} & \mathbf{B}_{21} & \mathbf{B}_{22} \end{bmatrix} \begin{bmatrix} y_{0t} \\ \mathbf{y}_{1t} \\ \mathbf{y}_{2t} \end{bmatrix} + \begin{bmatrix} \mathbf{\Gamma}_{01} & \mathbf{0}' \\ \mathbf{\Gamma}_{11} & \mathbf{\Gamma}_{12} \\ \mathbf{\Gamma}_{21} & \mathbf{\Gamma}_{22} \end{bmatrix} \begin{bmatrix} \mathbf{x}_{1t} \\ \mathbf{x}_{2t} \end{bmatrix} = \begin{bmatrix} u_{0t} \\ \mathbf{u}_{1t} \\ \mathbf{u}_{2t} \end{bmatrix}. \quad [9.3.5]$$

Here, for example, $\mathbf{B}_{12}$ is an $(n_1 \times n_2)$ matrix consisting of rows 2 through $(n_1 + 1)$ and columns $(n_1 + 2)$ through n of the matrix $\mathbf{B}$.

An alternative useful representation of the system is obtained by moving $\mathbf{\Gamma}\mathbf{x}_t$ to the right side of [9.3.1] and premultiplying both sides by $\mathbf{B}^{-1}$:

$$\mathbf{y}_t = -\mathbf{B}^{-1}\mathbf{\Gamma}\mathbf{x}_t + \mathbf{B}^{-1}\mathbf{u}_t = \mathbf{\Pi}'\mathbf{x}_t + \mathbf{v}_t, \quad [9.3.6]$$

where

$$\mathbf{\Pi}' = -\mathbf{B}^{-1}\mathbf{\Gamma} \quad [9.3.7]$$

$$\mathbf{v}_t = \mathbf{B}^{-1}\mathbf{u}_t. \quad [9.3.8]$$

Expression [9.3.6] is known as the *reduced-form* representation of the structural system [9.3.1]. In the reduced-form representation, each endogenous variable is expressed solely as a function of predetermined variables. For the example of [9.3.4], the reduced form is

$$\begin{aligned} \begin{bmatrix} q_t \\ p_t \end{bmatrix} &= -\begin{bmatrix} 1 & -\beta \\ 1 & -\gamma \end{bmatrix}^{-1} \begin{bmatrix} 0 \\ -h \end{bmatrix} w_t + \begin{bmatrix} 1 & -\beta \\ 1 & -\gamma \end{bmatrix}^{-1} \begin{bmatrix} u_t^d \\ u_t^s \end{bmatrix} \\ &= [1/(\beta - \gamma)] \begin{bmatrix} -\gamma & \beta \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ h \end{bmatrix} w_t \\ &\quad + [1/(\beta - \gamma)] \begin{bmatrix} -\gamma & \beta \\ -1 & 1 \end{bmatrix} \begin{bmatrix} u_t^d \\ u_t^s \end{bmatrix} \\ &= [1/(\beta - \gamma)] \begin{bmatrix} \beta h \\ h \end{bmatrix} w_t + [1/(\beta - \gamma)] \begin{bmatrix} -\gamma u_t^d + \beta u_t^s \\ -u_t^d + u_t^s \end{bmatrix}. \end{aligned} \quad [9.3.9]$$

The reduced form for a general system can be written in partitioned form as

$$\begin{bmatrix} y_{0t} \\ y_{1t} \\ y_{2t} \end{bmatrix} = \begin{bmatrix} \Pi_{01} & \Pi_{02} \\ \Pi_{11} & \Pi_{12} \\ \Pi_{21} & \Pi_{22} \end{bmatrix} \begin{bmatrix} x_{1t} \\ x_{2t} \end{bmatrix} + \begin{bmatrix} v_{0t} \\ v_{1t} \\ v_{2t} \end{bmatrix}, \quad [9.3.10]$$

where, for example, Π_{12} denotes an $(n_1 \times m_2)$ matrix consisting of rows 2 through $(n_1 + 1)$ and columns $(m_1 + 1)$ through m of the matrix Π' .

To apply the rank condition for identification of the first equation stated earlier, we would form the matrix of cross products between the explanatory variables in the first equation (x_{1t} and y_{1t}) and the predetermined variables for the whole system (x_{1t} and x_{2t}):

$$\mathbf{M} = \begin{bmatrix} E(\mathbf{x}_{1t} \mathbf{x}'_{1t}) & E(\mathbf{x}_{1t} \mathbf{x}'_{2t}) \\ E(\mathbf{y}_{1t} \mathbf{x}'_{1t}) & E(\mathbf{y}_{1t} \mathbf{x}'_{2t}) \end{bmatrix}. \quad [9.3.11]$$

In the earlier notation, the explanatory variables for the first equation consist of $\mathbf{z}_t = (\mathbf{x}'_{1t}, \mathbf{y}'_{1t})'$, while the predetermined variables for the system as a whole consist of $\mathbf{x}_t = (\mathbf{x}'_{1t}, \mathbf{x}'_{2t})'$. Thus, the rank condition, which required the rows of $E(\mathbf{z}_t \mathbf{x}'_t)$ to be linearly independent, amounts to the requirement that the rows of the $[(m_1 + n_1) \times m]$ matrix $\mathbf{M}$ in [9.3.11] be linearly independent. The rank condition can equivalently be stated in terms of the structural parameter matrices $\mathbf{B}$ and $\mathbf{\Gamma}$ or the reduced-form parameter matrix Π . The following proposition is adapted from Fisher (1966) and is proved in Appendix 9.A at the end of this chapter.

Proposition 9.1: *If the matrix $\mathbf{B}$ in [9.3.1] and the matrix of second moments of the predetermined variables $E(\mathbf{x}_t \mathbf{x}'_t)$ are both nonsingular, then the following conditions are equivalent:*

- (a) *The rows of the $[(m_1 + n_1) \times m]$ matrix $\mathbf{M}$ in [9.3.11] are linearly independent.*
- (b) *The rows of the $[(n_1 + n_2) \times (m_2 + n_2)]$ matrix*

$$\begin{bmatrix} \mathbf{\Gamma}_{12} & \mathbf{B}_{12} \\ \mathbf{\Gamma}_{22} & \mathbf{B}_{22} \end{bmatrix} \quad [9.3.12]$$

are linearly independent.

- (c) *The rows of the $(n_1 \times m_2)$ matrix Π_{12} are linearly independent.*

For example, for the system in [9.3.4], no endogenous variables are excluded from the first equation, and so $y_{0t} = q_t$, $y_{1t} = p_t$, and y_{2t} contains no elements. No predetermined variables appear in the first equation, and so x_{1t} contains no elements and $x_{2t} = w_t$. The matrix in [9.3.12] is then just given by the parameter $\mathbf{\Gamma}_{12}$. This represents the coefficient on x_{2t} in the equation describing y_{1t} and is equal to the scalar parameter $-h$. Result (b) of Proposition 9.1 thus states that the first equation is identified provided that $h \neq 0$. The value of Π_{12} can be read directly off the coefficient on w_t in the second row of [9.3.9] and turns out to be given by $h/(\beta - \gamma)$. Since $\mathbf{B}$ is assumed to be nonsingular, $(\beta - \gamma)$ is nonzero, and so $\mathbf{\Gamma}_{12}$ is zero if and only if Π_{12} is zero.

Achieving Identification Through Covariance Restrictions

Another way in which parameters can be identified is through restrictions on the covariances of the errors of the structural equations. For example, consider

again the supply and demand model, [9.3.2] and [9.3.3]. We saw that the demand elasticity β was identified by the exclusion of w_t from the demand equation. Consider now estimation of the supply elasticity γ .

Suppose first that we somehow knew the value of the demand elasticity β with certainty. Then the error in the demand equation could be constructed from

$$u_t^d = q_t - \beta p_t.$$

Notice that u_t^d would then be a valid instrument for the supply equation [9.3.3], since u_t^d is correlated with the endogenous explanatory variable for that equation (p_t) but u_t^d is uncorrelated with the error for that equation (u_t^s). Since w_t is also uncorrelated with the error u_t^s , it follows that the parameters of the supply equation could be estimated consistently by instrumental variable estimation with $\mathbf{x}_t = (u_t^d, w_t)'$:

$$\begin{bmatrix} \hat{\gamma}_T^* \\ \hat{h}_T^* \end{bmatrix} = \begin{bmatrix} \sum u_t^d p_t & \sum u_t^d w_t \\ \sum w_t p_t & \sum w_t^2 \end{bmatrix}^{-1} \begin{bmatrix} \sum u_t^d q_t \\ \sum w_t q_t \end{bmatrix} \xrightarrow{p} \begin{bmatrix} \gamma \\ h \end{bmatrix}, \quad [9.3.13]$$

where Σ indicates summation over $t = 1, 2, \dots, T$.

Although in practice we do not know the true value of β , it can be estimated consistently by IV estimation of [9.3.2] with w_t as an instrument:

$$\hat{\beta} = (\sum w_t p_t)^{-1} (\sum w_t q_t).$$

Then the residual u_t^d can be estimated consistently with $\hat{u}_t^d = q_t - \hat{\beta} p_t$. Consider, therefore, the estimator [9.3.13] with the population residual u_t^d replaced by the IV sample residual:

$$\begin{bmatrix} \hat{\gamma}_T \\ \hat{h}_T \end{bmatrix} = \begin{bmatrix} \sum \hat{u}_t^d p_t & \sum \hat{u}_t^d w_t \\ \sum w_t p_t & \sum w_t^2 \end{bmatrix}^{-1} \begin{bmatrix} \sum \hat{u}_t^d q_t \\ \sum w_t q_t \end{bmatrix}. \quad [9.3.14]$$

It is straightforward to use the fact that $\hat{\beta} \xrightarrow{p} \beta$ to deduce that the difference between the estimators in [9.3.14] and [9.3.13] converges in probability to zero. Hence, the estimator [9.3.14] is also consistent.

Two assumptions allowed the parameters of the supply equation (γ and h) to be estimated. First, an exclusion restriction allowed β to be estimated consistently. Second, a restriction on the covariance between u_t^d and u_t^s was necessary. If u_t^d were correlated with u_t^s , then u_t^d would not be a valid instrument for the supply equation and the estimator [9.3.13] would not be consistent.

Other Approaches to Identification

A good deal more can be said about identification. For example, parameters can also be identified through the imposition of certain restrictions on parameters such as $\beta_1 + \beta_2 = 1$. Useful references include Fisher (1966), Rothenberg (1971), and Hausman and Taylor (1983).

9.4. Full-Information Maximum Likelihood Estimation

Up to this point we have considered estimation of a single equation of the form $y_t = \beta' \mathbf{z}_t + u_t$. A more general approach is to specify a similar equation for every endogenous variable in the system, calculate the joint density of the vector of all of the endogenous variables conditional on the predetermined variables, and maximize the joint likelihood function. This is known as *full-information maximum likelihood* estimation, or *FIML*.

For illustration, suppose in [9.3.1] that the $(n \times 1)$ vector of structural disturbances $\mathbf{u}_t$ for date t is distributed $N(\mathbf{0}, \mathbf{D})$. Assume, further, that $\mathbf{u}_t$ is independent of $\mathbf{u}_\tau$ for $t \neq \tau$ and that $\mathbf{u}_t$ is independent of $\mathbf{x}_\tau$ for all t and τ . Then the reduced-form disturbance $\mathbf{v}_t = \mathbf{B}^{-1}\mathbf{u}_t$ is distributed $N(\mathbf{0}, \mathbf{B}^{-1}\mathbf{D}(\mathbf{B}^{-1})')$, and the reduced-form representation [9.3.6] implies that

$$\mathbf{y}_t|\mathbf{x}_t \sim N\left(\Pi'\mathbf{x}_t, \mathbf{B}^{-1}\mathbf{D}(\mathbf{B}^{-1})'\right) = N\left(-\mathbf{B}^{-1}\Gamma\mathbf{x}_t, \mathbf{B}^{-1}\mathbf{D}(\mathbf{B}^{-1})'\right).$$

The conditional log likelihood can then be found from

$$\begin{aligned}\mathcal{L}(\mathbf{B}, \Gamma, \mathbf{D}) &= \sum_{t=1}^T \log f(\mathbf{y}_t|\mathbf{x}_t; \mathbf{B}, \Gamma, \mathbf{D}) \\ &= -(Tn/2) \log(2\pi) - (T/2) \log|\mathbf{B}^{-1}\mathbf{D}(\mathbf{B}^{-1})'| \\ &\quad - (1/2) \sum_{t=1}^T [\mathbf{y}_t + \mathbf{B}^{-1}\Gamma\mathbf{x}_t]'[\mathbf{B}^{-1}\mathbf{D}(\mathbf{B}^{-1})']^{-1}[\mathbf{y}_t + \mathbf{B}^{-1}\Gamma\mathbf{x}_t].\end{aligned}\quad [9.4.1]$$

But

$$\begin{aligned}[\mathbf{y}_t + \mathbf{B}^{-1}\Gamma\mathbf{x}_t]'[\mathbf{B}^{-1}\mathbf{D}(\mathbf{B}^{-1})']^{-1}[\mathbf{y}_t + \mathbf{B}^{-1}\Gamma\mathbf{x}_t] &= [\mathbf{y}_t + \mathbf{B}^{-1}\Gamma\mathbf{x}_t]'[\mathbf{B}'\mathbf{D}^{-1}\mathbf{B}][\mathbf{y}_t + \mathbf{B}^{-1}\Gamma\mathbf{x}_t] \\ &= [\mathbf{B}(\mathbf{y}_t + \mathbf{B}^{-1}\Gamma\mathbf{x}_t)]'\mathbf{D}^{-1}[\mathbf{B}(\mathbf{y}_t + \mathbf{B}^{-1}\Gamma\mathbf{x}_t)] \\ &= [\mathbf{B}\mathbf{y}_t + \Gamma\mathbf{x}_t]'\mathbf{D}^{-1}[\mathbf{B}\mathbf{y}_t + \Gamma\mathbf{x}_t].\end{aligned}\quad [9.4.2]$$

Furthermore,

$$\begin{aligned}|\mathbf{B}^{-1}\mathbf{D}(\mathbf{B}^{-1})'| &= |\mathbf{B}^{-1}| \cdot |\mathbf{D}| \cdot |\mathbf{B}^{-1}| \\ &= |\mathbf{D}|/|\mathbf{B}|^2.\end{aligned}\quad [9.4.3]$$

Substituting [9.4.2] and [9.4.3] into [9.4.1],

$$\begin{aligned}\mathcal{L}(\mathbf{B}, \Gamma, \mathbf{D}) &= -(Tn/2) \log(2\pi) + (T/2) \log|\mathbf{B}|^2 \\ &\quad - (T/2) \log|\mathbf{D}| - (1/2) \sum_{t=1}^T [\mathbf{B}\mathbf{y}_t + \Gamma\mathbf{x}_t]'\mathbf{D}^{-1}[\mathbf{B}\mathbf{y}_t + \Gamma\mathbf{x}_t].\end{aligned}\quad [9.4.4]$$

The *FIML* estimates are then the values of $\mathbf{B}$, Γ , and $\mathbf{D}$ for which [9.4.4] is maximized.

For example, for the system of [9.3.4], the *FIML* estimates of β , γ , h , σ_d^2 , and σ_s^2 are found by maximizing

$$\begin{aligned}\mathcal{L}(\beta, \gamma, h, \sigma_d^2, \sigma_s^2) &= -T \log(2\pi) + \frac{T}{2} \log \begin{vmatrix} 1 & -\beta \\ 1 & -\gamma \end{vmatrix}^2 - \frac{T}{2} \log \begin{vmatrix} \sigma_d^2 & 0 \\ 0 & \sigma_s^2 \end{vmatrix} \\ &\quad - \frac{1}{2} \sum_{t=1}^T \left\{ \begin{bmatrix} q_t - \beta p_t & q_t - \gamma p_t - hw_t \end{bmatrix} \begin{bmatrix} \sigma_d^2 & 0 \\ 0 & \sigma_s^2 \end{bmatrix}^{-1} \begin{bmatrix} q_t - \beta p_t \\ q_t - \gamma p_t - hw_t \end{bmatrix} \right\} \\ &= -T \log(2\pi) + T \log(\gamma - \beta) - (T/2) \log(\sigma_d^2) \\ &\quad - (T/2) \log(\sigma_s^2) - (1/2) \sum_{t=1}^T (q_t - \beta p_t)^2 / \sigma_d^2 \\ &\quad - (1/2) \sum_{t=1}^T (q_t - \gamma p_t - hw_t)^2 / \sigma_s^2.\end{aligned}\quad [9.4.5]$$

The first-order conditions for maximization are

$$\frac{\partial \mathcal{L}}{\partial \beta} = -\frac{T}{\gamma - \beta} + \frac{\sum_{t=1}^T (q_t - \beta p_t)p_t}{\sigma_d^2} = 0 \quad [9.4.6]$$

$$\frac{\partial \mathcal{L}}{\partial \gamma} = \frac{T}{\gamma - \beta} + \frac{\sum_{t=1}^T (q_t - \gamma p_t - hw_t)p_t}{\sigma_s^2} = 0 \quad [9.4.7]$$

$$\frac{\partial \mathcal{L}}{\partial h} = \frac{\sum_{t=1}^T (q_t - \gamma p_t - hw_t)w_t}{\sigma_s^2} = 0 \quad [9.4.8]$$

$$\frac{\partial \mathcal{L}}{\partial \sigma_d^2} = -\frac{T}{2\sigma_d^2} + \frac{\sum_{t=1}^T (q_t - \beta p_t)^2}{2\sigma_d^4} = 0 \quad [9.4.9]$$

$$\frac{\partial \mathcal{L}}{\partial \sigma_s^2} = -\frac{T}{2\sigma_s^2} + \frac{\sum_{t=1}^T (q_t - \gamma p_t - hw_t)^2}{2\sigma_s^4} = 0. \quad [9.4.10]$$

The last two equations characterize the maximum likelihood estimates of the variances as the average squared residuals:

$$\hat{\sigma}_d^2 = (1/T) \sum_{t=1}^T (q_t - \hat{\beta} p_t)^2 \quad [9.4.11]$$

$$\hat{\sigma}_s^2 = (1/T) \sum_{t=1}^T (q_t - \hat{\gamma} p_t - \hat{h} w_t)^2. \quad [9.4.12]$$

Multiplying equation [9.4.7] by $(\beta - \gamma)/T$ results in

$$\begin{aligned} 0 &= -1 + \sum_{t=1}^T (q_t - \gamma p_t - hw_t)(\beta p_t - \gamma p_t)/(T\sigma_s^2) \\ &= -1 + \sum_{t=1}^T (q_t - \gamma p_t - hw_t)(\beta p_t - q_t + q_t - \gamma p_t)/(T\sigma_s^2). \end{aligned} \quad [9.4.13]$$

If [9.4.8] is multiplied by h/T and subtracted from [9.4.13], the result is

$$\begin{aligned} 0 &= -1 + \sum_{t=1}^T (q_t - \gamma p_t - hw_t)(\beta p_t - q_t + q_t - \gamma p_t - hw_t)/(T\sigma_s^2) \\ &= -1 + \sum_{t=1}^T (q_t - \gamma p_t - hw_t)(\beta p_t - q_t)/(T\sigma_s^2) \\ &\quad + \sum_{t=1}^T (q_t - \gamma p_t - hw_t)^2/(T\sigma_s^2) \\ &= -1 - \sum_{t=1}^T (q_t - \gamma p_t - hw_t)(q_t - \beta p_t)/(T\sigma_s^2) + 1, \end{aligned}$$

by virtue of [9.4.12]. Hence, the MLEs satisfy

$$\sum_{t=1}^T (q_t - \hat{\gamma} p_t - \hat{h} w_t)(q_t - \hat{\beta} p_t) = 0. \quad [9.4.14]$$

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Similarly, multiplying [9.4.6] by $(\gamma - \beta)/T$,

$$\begin{aligned} 0 &= -1 + \sum_{t=1}^T (q_t - \beta p_t)(\gamma p_t - q_t + q_t - \beta p_t)/(T\sigma_d^2) \\ &= -1 - \sum_{t=1}^T (q_t - \beta p_t)(q_t - \gamma p_t)/(T\sigma_d^2) + \sum_{t=1}^T (q_t - \beta p_t)^2/(T\sigma_d^2). \end{aligned}$$

Using [9.4.11],

$$\sum_{t=1}^T (q_t - \hat{\beta} p_t)(q_t - \hat{\gamma} p_t) = 0. \quad [9.4.15]$$

Subtracting [9.4.14] from [9.4.15],

$$0 = \sum_{t=1}^T (q_t - \hat{\beta} p_t)[(q_t - \hat{\gamma} p_t) - (q_t - \hat{\gamma} p_t - \hat{h} w_t)] = \hat{h} \sum_{t=1}^T (q_t - \hat{\beta} p_t) w_t.$$

Assuming that $\hat{h} \neq 0$, the *FIML* estimate of β thus satisfies

$$\sum_{t=1}^T (q_t - \hat{\beta} p_t) w_t = 0;$$

that is, the demand elasticity is chosen so as to make the estimated residual for the demand equation orthogonal to w_t . Hence, the instrumental variable estimator $\hat{\beta}_{IV}$ turns out also to be the *FIML* estimator. Equations [9.4.8] and [9.4.14] assert that the parameters for the supply equation (γ and h) are chosen so as to make the residual for that equation orthogonal to w_t and to the demand residual $\hat{u}_t^d = q_t - \hat{\beta} p_t$. Hence, the *FIML* estimates for these parameters are the same as the instrumental-variable estimates suggested in [9.3.14].

For this example, two-stage least squares, instrumental variable estimation, and full-information maximum likelihood all produced the identical estimates. This is because the model is *just identified*. A model is said to be just identified if for any admissible value for the parameters of the reduced-form representation there exists a unique value for the structural parameters that implies those reduced-form parameters. A model is said to be *overidentified* if some admissible values for the reduced-form parameters are ruled out by the structural restrictions. In an overidentified model, *IV*, *2SLS*, and *FIML* estimation are not equivalent, and *FIML* typically produces the most efficient estimates.

For a general overidentified simultaneous equation system with no restrictions on the variance-covariance matrix, the *FIML* estimates can be calculated by iterating on a procedure known as *three-stage least squares*; see, for example, Maddala (1977, pp. 482–90). Rothenberg and Ruud (1990) discussed *FIML* estimation in the presence of covariance restrictions. *FIML* estimation of dynamic time series models will be discussed further in Chapter 11.

9.5 Estimation Based on the Reduced Form

If a system is just identified as in [9.3.2] and [9.3.3] with u_t^d uncorrelated with u_t^s , one approach is to maximize the likelihood function with respect to the reduced-form parameters. The values of the structural parameters associated with these values for the reduced-form parameters are the same as the *FIML* estimates in a just-identified model.

The log likelihood [9.4.1] can be expressed in terms of the reduced-form parameters $\mathbf{\Pi}$ and $\mathbf{\Omega}$ as

$$\begin{aligned}\mathcal{L}(\mathbf{\Pi}, \mathbf{\Omega}) &= \sum_{t=1}^T \log f(\mathbf{y}_t | \mathbf{x}_t; \mathbf{\Pi}, \mathbf{\Omega}) \\ &= -(Tn/2) \log(2\pi) - (T/2) \log |\mathbf{\Omega}| \\ &\quad - (1/2) \sum_{t=1}^T [\mathbf{y}_t - \mathbf{\Pi}' \mathbf{x}_t]' \mathbf{\Omega}^{-1} [\mathbf{y}_t - \mathbf{\Pi}' \mathbf{x}_t],\end{aligned}\tag{9.5.1}$$

where $\mathbf{\Omega} = E(\mathbf{v}_t \mathbf{v}_t') = \mathbf{B}^{-1} \mathbf{D} (\mathbf{B}^{-1})'$. The value of $\mathbf{\Pi}$ that maximizes [9.5.1] will be shown in Chapter 11 to be given by

$$\hat{\mathbf{\Pi}}' = \left[\sum_{t=1}^T \mathbf{y}_t \mathbf{x}_t' \right] \left[\sum_{t=1}^T \mathbf{x}_t \mathbf{x}_t' \right]^{-1};$$

in other words, the i th row of $\hat{\mathbf{\Pi}}'$ is obtained from an *OLS* regression of the i th endogenous variable on all of the predetermined variables:

$$\hat{\pi}_i' = \left[\sum_{t=1}^T y_{it} \mathbf{x}_t' \right] \left[\sum_{t=1}^T \mathbf{x}_t \mathbf{x}_t' \right]^{-1}.$$

The *MLE* of $\mathbf{\Omega}$ turns out to be

$$\hat{\mathbf{\Omega}} = (1/T) \left[\sum_{t=1}^T (\mathbf{y}_t - \hat{\mathbf{\Pi}}' \mathbf{x}_t)(\mathbf{y}_t - \hat{\mathbf{\Pi}}' \mathbf{x}_t)' \right].$$

For a just-identified model, the *FIML* estimates are the values of $(\mathbf{B}, \mathbf{\Gamma}, \mathbf{D})$ for which $\hat{\mathbf{\Pi}}' = -\mathbf{B}^{-1} \mathbf{\Gamma}$ and $\hat{\mathbf{\Omega}} = \mathbf{B}^{-1} \mathbf{D} (\mathbf{B}^{-1})'$.

We now show that the estimates of $\mathbf{B}$, $\mathbf{\Gamma}$, and $\mathbf{D}$ inferred in this fashion from the reduced-form parameters for the just-identified supply-and-demand example are the same as the *FIML* estimates. The estimate $\hat{\pi}_1$ is found by *OLS* regression of q_t on w_t , while $\hat{\pi}_2$ is the coefficient from an *OLS* regression of p_t on w_t . These estimates satisfy

$$\sum_{t=1}^T (q_t - \hat{\pi}_1 w_t) w_t = 0 \tag{9.5.2}$$

$$\sum_{t=1}^T (p_t - \hat{\pi}_2 w_t) w_t = 0 \tag{9.5.3}$$

and

$$\begin{bmatrix} \hat{\Omega}_{11} & \hat{\Omega}_{12} \\ \hat{\Omega}_{21} & \hat{\Omega}_{22} \end{bmatrix} = (1/T) \begin{bmatrix} \Sigma(q_t - \hat{\pi}_1 w_t)^2 & \Sigma(q_t - \hat{\pi}_1 w_t)(p_t - \hat{\pi}_2 w_t) \\ \Sigma(p_t - \hat{\pi}_2 w_t)(q_t - \hat{\pi}_1 w_t) & \Sigma(p_t - \hat{\pi}_2 w_t)^2 \end{bmatrix}.\tag{9.5.4}$$

The structural estimates satisfy $\mathbf{B} \hat{\mathbf{\Pi}}' = -\mathbf{\Gamma}$ or

$$\begin{bmatrix} 1 & -\beta \\ 1 & -\gamma \end{bmatrix} \begin{bmatrix} \hat{\pi}_1 \\ \hat{\pi}_2 \end{bmatrix} = \begin{bmatrix} 0 \\ h \end{bmatrix}.\tag{9.5.5}$$

Multiplying [9.5.3] by β and subtracting the result from [9.5.2] produces

$$\begin{aligned} 0 &= \sum_{t=1}^T (q_t - \hat{\pi}_1 w_t - \beta p_t + \beta \hat{\pi}_2 w_t) w_t \\ &= \sum_{t=1}^T (q_t - \beta p_t) w_t - \sum_{t=1}^T (\hat{\pi}_1 - \beta \hat{\pi}_2) w_t^2 \\ &= \sum_{t=1}^T (q_t - \beta p_t) w_t, \end{aligned}$$

by virtue of the first row of [9.5.5]. Thus, the estimate of β inferred from the reduced-form parameters is the same as the *IV* or *FIML* estimate derived earlier. Similarly, multiplying [9.5.3] by γ and subtracting the result from [9.5.2] gives

$$\begin{aligned} 0 &= \sum_{t=1}^T (q_t - \hat{\pi}_1 w_t - \gamma p_t + \gamma \hat{\pi}_2 w_t) w_t \\ &= \sum_{t=1}^T [q_t - \gamma p_t - (\hat{\pi}_1 - \gamma \hat{\pi}_2) w_t] w_t \\ &= \sum_{t=1}^T [q_t - \gamma p_t - h w_t] w_t, \end{aligned}$$

by virtue of the second row of [9.5.5], reproducing the first-order condition [9.4.8] for *FIML*. Finally, we need to solve $\mathbf{D} = \mathbf{B}\hat{\Omega}\mathbf{B}'$ for $\mathbf{D}$ and γ (the remaining element of $\mathbf{B}$). These equations are

$$\begin{aligned} \begin{bmatrix} \sigma_d^2 & 0 \\ 0 & \sigma_s^2 \end{bmatrix} &= \begin{bmatrix} 1 & -\beta \\ 1 & -\gamma \end{bmatrix} \begin{bmatrix} \hat{\Omega}_{11} & \hat{\Omega}_{12} \\ \hat{\Omega}_{21} & \hat{\Omega}_{22} \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -\beta & -\gamma \end{bmatrix} \\ &= \frac{1}{T} \sum_{t=1}^T \left\{ \begin{bmatrix} 1 & -\beta \\ 1 & -\gamma \end{bmatrix} \begin{bmatrix} q_t - \hat{\pi}_1 w_t \\ p_t - \hat{\pi}_2 w_t \end{bmatrix} \begin{bmatrix} q_t - \hat{\pi}_1 w_t & p_t - \hat{\pi}_2 w_t \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -\beta & -\gamma \end{bmatrix} \right\} \\ &= \frac{1}{T} \sum_{t=1}^T \left\{ \begin{bmatrix} q_t - \beta p_t - (\hat{\pi}_1 - \beta \hat{\pi}_2) w_t \\ q_t - \gamma p_t - (\hat{\pi}_1 - \gamma \hat{\pi}_2) w_t \end{bmatrix} \begin{bmatrix} q_t - \beta p_t - (\hat{\pi}_1 - \beta \hat{\pi}_2) w_t & q_t - \gamma p_t - (\hat{\pi}_1 - \gamma \hat{\pi}_2) w_t \end{bmatrix}' \right\} \\ &= \frac{1}{T} \sum_{t=1}^T \left\{ \begin{bmatrix} q_t - \beta p_t \\ q_t - \gamma p_t - h w_t \end{bmatrix} \begin{bmatrix} q_t - \beta p_t & q_t - \gamma p_t - h w_t \end{bmatrix} \right\}. \end{aligned}$$

The diagonal elements of this matrix system of equations reproduce the earlier formulas for the *FIML* estimates of the variance parameters, while the off-diagonal element reproduces the result [9.4.14].

9.6. Overview of Simultaneous Equations Bias

The problem of simultaneous equations bias is extremely widespread in the social sciences. It is rare that the relation that we are interested in estimating is the only possible reason why the dependent and explanatory variables might be correlated. For example, consider trying to estimate the effect of military service on an individual's subsequent income. This parameter cannot be estimated by a regression of income on a measure of military service and other observed variables. The error

term in such a regression represents other characteristics of the individual that influence income, and these omitted factors are also likely to have influenced the individual's military participation. As another example, consider trying to estimate the success of long prison sentences in deterring crime. This cannot be estimated by regressing the crime rate in a state on the average prison term in that state, because some states may have adopted stiffer prison sentences in response to higher crime. The error term in the regression, which represents other factors that influence crime, is thus likely also to be correlated with the explanatory variable. Regardless of whether the researcher is interested in the factors that determine military service or prison terms or has any theory about them, simultaneous equations bias must be recognized and dealt with.

Furthermore, it is not enough to find an instrument $\mathbf{x}_t$ that is uncorrelated with the residual u_t . In order to satisfy the rank condition, the instrument $\mathbf{x}_t$ must be correlated with the endogenous explanatory variables $\mathbf{z}_t$. The calculations by Nelson and Startz (1990) suggest that very poor estimates can result if $\mathbf{x}_t$ is only weakly correlated with $\mathbf{z}_t$.

Finding valid instruments is often extremely difficult and requires careful thought and a bit of good luck. For the question about military service, Angrist (1990) found an ingenious instrument for military service based on the institutional details of the draft in the United States during the Vietnam War. The likelihood that an individual was drafted into military service was determined by a lottery based on birthdays. Thus, an individual's birthday during the year would be correlated with military service but presumably uncorrelated with other factors influencing income. Unfortunately, it is unusual to be able to find such a compelling instrument for many questions that one would like to ask of the data.

APPENDIX 9.A. *Proofs of Chapter 9 Proposition*

■ **Proof of Proposition 9.1.** We first show that (a) implies (c). The middle block of [9.3.10] states that

$$\mathbf{y}_{1t} = \Pi_{11}\mathbf{x}_{1t} + \Pi_{12}\mathbf{x}_{2t} + \mathbf{v}_{1t}.$$

Hence,

$$\begin{aligned} \mathbf{M} &= E\left\{\begin{bmatrix} \mathbf{x}_{1t} \\ \mathbf{y}_{1t} \end{bmatrix} \begin{bmatrix} \mathbf{x}'_{1t} & \mathbf{x}'_{2t} \end{bmatrix}\right\} \\ &= E\left\{\begin{bmatrix} \mathbf{I}_{m_1} & \mathbf{0} \\ \Pi_{11} & \Pi_{12} \end{bmatrix} \begin{bmatrix} \mathbf{x}_{1t} \\ \mathbf{x}_{2t} \end{bmatrix} \begin{bmatrix} \mathbf{x}'_{1t} & \mathbf{x}'_{2t} \end{bmatrix} + \begin{bmatrix} \mathbf{0} \\ \mathbf{v}_{1t} \end{bmatrix} \begin{bmatrix} \mathbf{x}'_{1t} & \mathbf{x}'_{2t} \end{bmatrix}\right\} \quad [9.A.1] \\ &= \begin{bmatrix} \mathbf{I}_{m_1} & \mathbf{0} \\ \Pi_{11} & \Pi_{12} \end{bmatrix} E(\mathbf{x}_t \mathbf{x}'_t), \end{aligned}$$

since $\mathbf{x}_t$ is uncorrelated with $\mathbf{u}_t$ and thus uncorrelated with $\mathbf{v}_{1t}$.

Suppose that the rows of $\mathbf{M}$ are linearly independent. This means that $[\boldsymbol{\lambda}' \quad \boldsymbol{\mu}']\mathbf{M} \neq \mathbf{0}'$ for any $(m_1 \times 1)$ vector $\boldsymbol{\lambda}$ and any $(n_1 \times 1)$ vector $\boldsymbol{\mu}$ that are not both zero. In particular, $[-\boldsymbol{\mu}'\Pi_{11} \quad \boldsymbol{\mu}']\mathbf{M} \neq \mathbf{0}'$. But from the right side of [9.A.1], this implies that

$$[-\boldsymbol{\mu}'\Pi_{11} \quad \boldsymbol{\mu}'] \begin{bmatrix} \mathbf{I}_{m_1} & \mathbf{0} \\ \Pi_{11} & \Pi_{12} \end{bmatrix} E(\mathbf{x}_t \mathbf{x}'_t) = [\mathbf{0}' \quad \boldsymbol{\mu}'\Pi_{12}] E(\mathbf{x}_t \mathbf{x}'_t) \neq \mathbf{0}'$$

for any nonzero $(n_1 \times 1)$ vector $\boldsymbol{\mu}$. But this could be true only if $\boldsymbol{\mu}'\Pi_{12} \neq \mathbf{0}'$. Hence, if the rows of $\mathbf{M}$ are linearly independent, then the rows of Π_{12} are also linearly independent.

To prove that (c) implies (a), premultiply both sides of [9.A.1] by any nonzero vector $[\lambda' \quad \mu']$. The right side becomes

$$[\lambda' \quad \mu'] \begin{bmatrix} \mathbf{I}_{m_1} & \mathbf{0} \\ \mathbf{\Pi}_{11} & \mathbf{\Pi}_{12} \end{bmatrix} E(\mathbf{x}_i, \mathbf{x}_i') = [(\lambda' + \mu' \mathbf{\Pi}_{11}) \quad \mu' \mathbf{\Pi}_{12}] E(\mathbf{x}_i, \mathbf{x}_i') \equiv \boldsymbol{\eta}' E(\mathbf{x}_i, \mathbf{x}_i'),$$

where $\boldsymbol{\eta}' \equiv [(\lambda' + \mu' \mathbf{\Pi}_{11}) \quad \mu' \mathbf{\Pi}_{12}]$. If the rows of $\mathbf{\Pi}_{12}$ are linearly independent, then $\boldsymbol{\eta}'$ cannot be the zero vector unless both μ and λ are zero. To see this, note that if μ is nonzero, then $\mu' \mathbf{\Pi}_{12}$ cannot be the zero vector, while if $\mu = \mathbf{0}$, then $\boldsymbol{\eta}$ will be zero only if λ is also the zero vector. Furthermore, since $E(\mathbf{x}_i, \mathbf{x}_i')$ is nonsingular, a nonzero $\boldsymbol{\eta}$ means that $\boldsymbol{\eta}' E(\mathbf{x}_i, \mathbf{x}_i')$ cannot be the zero vector. Thus, if the right side of [9.A.1] is premultiplied by any nonzero vector (λ', μ') , the result is not zero. The same must be true of the left side: $[\lambda' \quad \mu'] \mathbf{M} \neq \mathbf{0}'$ for any nonzero (λ', μ') , establishing that linear independence of the rows of $\mathbf{\Pi}_{12}$ implies linear independence of the rows of $\mathbf{M}$.

To see that (b) implies (c), write [9.3.7] as

$$\begin{bmatrix} \mathbf{\Pi}_{01} & \mathbf{\Pi}_{02} \\ \mathbf{\Pi}_{11} & \mathbf{\Pi}_{12} \\ \mathbf{\Pi}_{21} & \mathbf{\Pi}_{22} \end{bmatrix} = -\mathbf{B}^{-1} \begin{bmatrix} \mathbf{\Gamma}_{01} & \mathbf{0}' \\ \mathbf{\Gamma}_{11} & \mathbf{\Gamma}_{12} \\ \mathbf{\Gamma}_{21} & \mathbf{\Gamma}_{22} \end{bmatrix}. \quad [9.A.2]$$

We also have the identity

$$\begin{bmatrix} 1 & \mathbf{0}' & \mathbf{0}' \\ \mathbf{0} & \mathbf{I}_{n_1} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{I}_{n_2} \end{bmatrix} = \mathbf{B}^{-1} \begin{bmatrix} 1 & \mathbf{B}_{01} & \mathbf{0}' \\ \mathbf{B}_{10} & \mathbf{B}_{11} & \mathbf{B}_{12} \\ \mathbf{B}_{20} & \mathbf{B}_{21} & \mathbf{B}_{22} \end{bmatrix}. \quad [9.A.3]$$

The system of equations represented by the second block column of [9.A.2] and the third block column of [9.A.3] can be collected as

$$\begin{bmatrix} \mathbf{\Pi}_{02} & \mathbf{0}' \\ \mathbf{\Pi}_{12} & \mathbf{0} \\ \mathbf{\Pi}_{22} & \mathbf{I}_{n_2} \end{bmatrix} = \mathbf{B}^{-1} \begin{bmatrix} \mathbf{0}' & \mathbf{0}' \\ -\mathbf{\Gamma}_{12} & \mathbf{B}_{12} \\ -\mathbf{\Gamma}_{22} & \mathbf{B}_{22} \end{bmatrix}. \quad [9.A.4]$$

If both sides of [9.A.4] are premultiplied by the row vector $[0 \quad \mu_1' \quad \mathbf{0}']$ where μ_1 is any $(n_1 \times 1)$ vector, the result is

$$\begin{aligned} [\mu_1' \mathbf{\Pi}_{12} \quad \mathbf{0}'] &= [0 \quad \mu_1' \quad \mathbf{0}'] \mathbf{B}^{-1} \begin{bmatrix} \mathbf{0}' & \mathbf{0}' \\ -\mathbf{\Gamma}_{12} & \mathbf{B}_{12} \\ -\mathbf{\Gamma}_{22} & \mathbf{B}_{22} \end{bmatrix} \\ &= [\lambda_0 \quad \lambda_1' \quad \lambda_2'] \begin{bmatrix} \mathbf{0}' & \mathbf{0}' \\ -\mathbf{\Gamma}_{12} & \mathbf{B}_{12} \\ -\mathbf{\Gamma}_{22} & \mathbf{B}_{22} \end{bmatrix} \\ &= [\lambda_1' \quad \lambda_2'] \begin{bmatrix} -\mathbf{\Gamma}_{12} & \mathbf{B}_{12} \\ -\mathbf{\Gamma}_{22} & \mathbf{B}_{22} \end{bmatrix}, \end{aligned} \quad [9.A.5]$$

where

$$[\lambda_0 \quad \lambda_1' \quad \lambda_2'] \equiv [0 \quad \mu_1' \quad \mathbf{0}'] \mathbf{B}^{-1},$$

implying

$$[0 \quad \mu_1' \quad \mathbf{0}'] = [\lambda_0 \quad \lambda_1' \quad \lambda_2'] \mathbf{B}. \quad [9.A.6]$$

Suppose that the rows of the matrix $\begin{bmatrix} \mathbf{\Gamma}_{12} & \mathbf{B}_{12} \\ \mathbf{\Gamma}_{22} & \mathbf{B}_{22} \end{bmatrix}$ are linearly independent. Then the only values for λ_1 and λ_2 for which the right side of [9.A.5] can be zero are $\lambda_1 = \mathbf{0}$ and $\lambda_2 = \mathbf{0}$. Substituting these values into [9.A.6], the only value of μ_1 for which the left side of [9.A.5] can be zero must satisfy

$$\begin{aligned} [0 \quad \mu_1' \quad \mathbf{0}'] &= [\lambda_0 \quad \mathbf{0}' \quad \mathbf{0}'] \mathbf{B} \\ &= [\lambda_0 \quad \lambda_0 \mathbf{B}_{01} \quad \mathbf{0}']. \end{aligned}$$

Matching the first elements in these vectors implies $\lambda_0 = 0$, and thus matching the second elements requires $\mu_1 = \mathbf{0}$. Thus, if condition (b) is satisfied, then the only value of μ_1 for

which the left side of [9.A.5] can be zero is $\mu_1 = \mathbf{0}$, establishing that the rows of Π_{12} are linearly independent. Hence, condition (c) is satisfied whenever (b) holds.

Conversely, to see that (c) implies (b), let λ_1 and λ_2 denote any $(n_1 \times 1)$ and $(n_2 \times 1)$ vectors, and premultiply both sides of [9.A.4] by the row vector $[0 \quad \lambda_1' \quad \lambda_2']\mathbf{B}$:

$$[0 \quad \lambda_1' \quad \lambda_2']\mathbf{B} \begin{bmatrix} \Pi_{02} & \mathbf{0}' \\ \Pi_{12} & \mathbf{0} \\ \Pi_{22} & \mathbf{I}_{n_2} \end{bmatrix} = [0 \quad \lambda_1' \quad \lambda_2'] \begin{bmatrix} \mathbf{0}' & \mathbf{0}' \\ -\Gamma_{12} & \mathbf{B}_{12} \\ -\Gamma_{22} & \mathbf{B}_{22} \end{bmatrix}$$

or

$$[\mu_0 \quad \mu_1' \quad \mu_2'] \begin{bmatrix} \Pi_{02} & \mathbf{0}' \\ \Pi_{12} & \mathbf{0} \\ \Pi_{22} & \mathbf{I}_{n_2} \end{bmatrix} = [\lambda_1' \quad \lambda_2'] \begin{bmatrix} -\Gamma_{12} & \mathbf{B}_{12} \\ -\Gamma_{22} & \mathbf{B}_{22} \end{bmatrix} \quad [9.A.7]$$

where

$$[\mu_0 \quad \mu_1' \quad \mu_2'] \equiv [0 \quad \lambda_1' \quad \lambda_2']\mathbf{B}. \quad [9.A.8]$$

Premultiplying both sides of equation [9.A.4] by $\mathbf{B}$ implies that

$$\begin{bmatrix} 1 & \mathbf{B}_{01} & \mathbf{0}' \\ \mathbf{B}_{10} & \mathbf{B}_{11} & \mathbf{B}_{12} \\ \mathbf{B}_{20} & \mathbf{B}_{21} & \mathbf{B}_{22} \end{bmatrix} \begin{bmatrix} \Pi_{02} & \mathbf{0}' \\ \Pi_{12} & \mathbf{0} \\ \Pi_{22} & \mathbf{I}_{n_2} \end{bmatrix} = \begin{bmatrix} \mathbf{0}' & \mathbf{0}' \\ -\Gamma_{12} & \mathbf{B}_{12} \\ -\Gamma_{22} & \mathbf{B}_{22} \end{bmatrix}.$$

The upper left element of this matrix system asserts that

$$\Pi_{02} + \mathbf{B}_{01}\Pi_{12} = \mathbf{0}'. \quad [9.A.9]$$

Substituting [9.A.9] into [9.A.7],

$$[\mu_0 \quad \mu_1' \quad \mu_2'] \begin{bmatrix} -\mathbf{B}_{01}\Pi_{12} & \mathbf{0}' \\ \Pi_{12} & \mathbf{0} \\ \Pi_{22} & \mathbf{I}_{n_2} \end{bmatrix} = [\lambda_1' \quad \lambda_2'] \begin{bmatrix} -\Gamma_{12} & \mathbf{B}_{12} \\ -\Gamma_{22} & \mathbf{B}_{22} \end{bmatrix}. \quad [9.A.10]$$

In order for the left side of [9.A.10] to be zero, it must be the case that $\mu_2 = \mathbf{0}$ and that

$$-\mu_0\mathbf{B}_{01}\Pi_{12} + \mu_1'\Pi_{12} = (\mu_1' - \mu_0\mathbf{B}_{01})\Pi_{12} = \mathbf{0}'. \quad [9.A.11]$$

But if the rows of Π_{12} are linearly independent, [9.A.11] can be zero only if

$$\mu_1' = \mu_0\mathbf{B}_{01}. \quad [9.A.12]$$

Substituting these results into [9.A.8], it follows that [9.A.10] can be zero only if

$$\begin{aligned} [0 \quad \lambda_1' \quad \lambda_2']\mathbf{B} &= [\mu_0 \quad \mu_0\mathbf{B}_{01} \quad \mathbf{0}'] \\ &= [\mu_0 \quad \mathbf{0}' \quad \mathbf{0}'] \begin{bmatrix} 1 & \mathbf{B}_{01} & \mathbf{0}' \\ \mathbf{B}_{10} & \mathbf{B}_{11} & \mathbf{B}_{12} \\ \mathbf{B}_{20} & \mathbf{B}_{21} & \mathbf{B}_{22} \end{bmatrix} \\ &= [\mu_0 \quad \mathbf{0}' \quad \mathbf{0}']\mathbf{B}. \end{aligned} \quad [9.A.13]$$

Since $\mathbf{B}$ is nonsingular, both sides of [9.A.13] can be postmultiplied by $\mathbf{B}^{-1}$ to deduce that [9.A.10] can be zero only if

$$[0 \quad \lambda_1' \quad \lambda_2'] = [\mu_0 \quad \mathbf{0}' \quad \mathbf{0}'].$$

Thus, the right side of [9.A.10] can be zero only if λ_1 and λ_2 are both zero, establishing that the rows of the matrix in [9.3.12] must be linearly independent. ■

Chapter 9 Exercise

9.1. Verify that [9.2.23] gives a consistent estimate of σ^2 .

Chapter 9 References

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